IQP Generative Models for Strong Gravitational Lensing Images

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Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed hard to sample classically [5, 6], ~~while~~ yet the expectation values of a maximum-mean-discrepancy (MMD) loss ~~needs~~ stay computable, so ~~such a~~ an IQP Born machine can be trained classically and deployed on quantum hardware [24]. A recent correlation–complexity map is proposed to indicate which data suit ~~them~~such models, and DeepLense strong-lensing images sit at ~~the~~ its most IQP-compatible position among the encodings we measured. ~~The reference protocol's frozen core with sequential adaptation yields only noise, which joint training repairs , and a learned linear~~ On these images the reference protocol, a core frozen on one anchor followed by sequential per-image adaptation, generates noise. Joint training repairs it. A learned linear latent map then loses coherence past $M{=}6$, so ~~we let~~ a small non-linear map ~~carry the latent . At~~ carries the latent instead. With eleven training images the resulting 10-qubit machine matches a DCGAN ~~of~~ with six times as many parameters on both generation metrics within the seed spread~~, is ahead of it~~ . It leads the DCGAN on a flux-aligned variant~~and of~~ , leads an RBM on the two MMD^2 columns~~while level with it~~ , is level with the RBM on the histogram metric, and projects unseen images at correlation 0.901~~; at~~ . With 100 images the DCGAN ~~is ahead and that~~ leads and the correlation reaches 0.977. ~~Every circuit is~~ All circuits are simulated exactly at $n = 10$ qubits over five seeds, so we claim no quantum advantage ~~, only~~ and treat the work as an early test of whether such a model could be worth deploying where training data are scarce.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard [30] transform of a phase function. Exact or multiplicatively approximate classical sampling from them would collapse the polynomial hierarchy [5], and ~~that~~ the hardness is conjectured to hold on average under additive error [6]~~, a bounded motivation , since noise~~ . The motivation has limits. Noise makes such circuits simulable in polynomial time [23], classical algorithms keep improving [18], and trainability itself pulls towards simulability [7]. The same structure keeps the expectation values of an MMD loss ~~'s expectation values~~ computable [25], as ~~it does~~ for other encodings built to stay simulable [17]~~, so such~~ . Such a model can ~~be *trained classically and deployed on quantum hardware*~~therefore be trained classically and deployed on quantum hardware [24], ~~in the line of~~ following earlier Born machines trained ~~by~~ with an MMD loss [16, 3, 9].

Whether real data sit where ~~that is useful is empirical , and a~~ this helps is an empirical question. A recent proposal [15] approaches it with ~~two ingredients. (i) A~~ a *correlation–complexity map*~~placing~~ . The map places a dataset at I_{QCLI}, the Jensen–Shannon divergence between its Walsh order spectrum ~~, (~~the Fourier power at each Hamming weight~~,)~~ and the i.i.d. binomial baseline, and at

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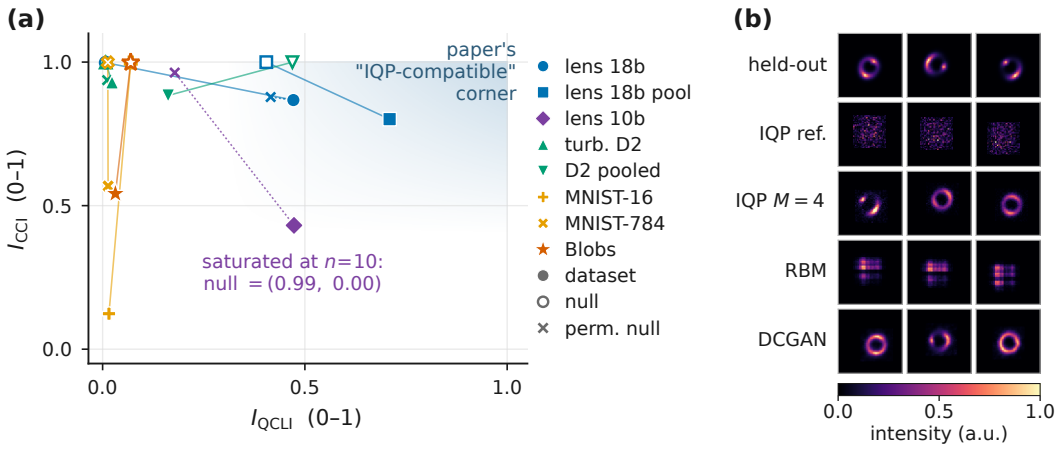


Figure 1: **(a)** Correlation–complexity map ~~,~~ with datasets filled, i.i.d.-uniform nulls open ~~,~~ and our permutation nulls as crosses~~(ours); lens~~. Lens 18b and 18b pool are the per-image and pooled 18-bit encodings, lens 10b is the crop pmf ~~,~~ and turb. D2 is the reference turbulence data. **(b)** ~~First~~ The first three unconditional samples per model ~~,~~ at $N{=}100$, ~~no~~ without selection.

$I_{\mathrm{CCI}} \in [0, 1]$, the share of its correlation that the best pairwise tree misses [31, 26, 8]~~,~~. IQP circuits should do well in the high-I_{QCLI}, high-I_{CCI} corner~~being where IQP circuits should do well. (ii) A~~. The same paper gives a latent-adaptation protocol (Algorithm 1 of ~~that paper's~~ its v2~~,~~), in which a phase core is trained on one anchor ~~plus~~ and a block of the first 50 phases in Hamming order ~~,~~ is randomly initialised and adapted sample by sample~~),~~. It was demonstrated at $n = 18$ qubits on 2D turbulence [13], one of several datasets the map covers. A follow-up initialises ~~that~~ the same fixed block from a physics-informed surrogate [14], ~~where we instead~~ whereas we learn the directions.

We ~~take~~ apply both to simulated dark-matter strong-lensing images from DeepLense [1, 2]~~and contribute three things. We recompute the map on lens crops , which~~. Lens crops land at the most IQP-compatible map position we measured, and ~~show with~~ a permutation null ~~that there the index is largely determined~~ shows that their I_{QCLI} is largely set by the intensity histogram, so ~~a favourable~~ that position is not by itself evidence of IQP-suitable structure. ~~We trace why~~ On static images the reference latent protocol produces noise~~on static images~~. We trace the cause and replace its latent map by a non-linear one~~. We then compare against~~, then compare the model with an RBM and a DCGAN under two intensity conventions~~, and report~~. App. C reports a replication on the turbulence data~~in App. C.~~.

2 Setup

~~**Model.**~~ Four choices differ from the reference setup, each following from the data. The reference snapshots interpolate to unseen times, while the DeepLense `Model II` images we use [19] are static, so we generate unconditionally and project onto held-out images. Its 18-bit $(x, y, \text{intensity})$ encoding suits a field of tuples, but the raw intensity axis of a lens crop carries barely one bit. We instead centroid-align each image and read its central 32×32 crop, which holds 99.4% of the light, as one probability mass function (pmf) over 1,024 cells at $n = 10$ (App. A). The reference fixes the latent block, whereas we learn the map and train the N images jointly. It also estimates its MMD from samples, while a 10-qubit pmf is exact.

Every arm is a 10-qubit IQP Born machine over 1,024 Walsh phases θ, one per qubit subset~~and 1,023 of them effective since the~~. The empty subset is a global phase, ~~differing~~ which leaves 1,023 effective. The arms differ only in how a per-image latent enters. The reference protocol freezes the first 50 ~~of those~~ phases in Hamming order and adapts them sample by sample~~, which on static crops leaves~~. On static crops this reconstructs the other training images poorly ~~reconstructed~~ (Sec. 3.2), so we make ~~that~~ the map trainable, first as a linear map of M directions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with each row of Φ normalised in L^2. ~~That~~ On these crops the linear map degrades by $M{=}6$~~on these crops, we suspect once~~. We suspect the $\|z\|$ the images demand ~~leaves~~ exceeds the radius over which the Walsh phases interfere coherently, ~~though we do not isolate the cause; our~~ but we have not isolated the cause. The main arm replaces $\Phi^\top z$ by a one-hidden-layer $\tanh$ map g of width $H{=}32$, with the z_i trained jointly with g under a small L^2 penalty. ~~The main~~ Its objective is the flat MMD (uniform Walsh weights, estimable from hardware shots)~~, with the~~. The three-bandwidth Gaussian kernel of [24], the *paper kernel* of the tables, is kept as a secondary arm~~, and~~. θ^0 is warm-started by 1,000 exact-KL steps~~, which a flat-MMD warm start replaces at no cost (App. C).~~.

~~Decoding, baselines and metrics.~~ The trained pmf fixes only the relative intensities~~, so it is pasted~~. We paste it into a 64×64 frame at a total intensity S and at an offset drawn from the ~~training crops' own offsets~~offsets of the training crops. The reference convention gives every frame the training-set mean S, ~~leaving~~ so the generated flux ~~with zero variance where~~ has zero variance while real crops spread by about a quarter of their mean~~, so we~~. We therefore draw S per image from a kernel density estimate of the N training totals, with no retraining~~;~~, and App. A repeats everything under the fixed S. The baselines are a restricted Boltzmann machine (RBM) [28, 12, 29] and a deep convolutional generative adversarial network (DCGAN) [22]~~, both~~. Both are early-stopped on 200 validation images, while the IQP arms run a fixed 4,000 steps, which can only favour ~~them. We score with~~ the baselines. PDF–JS is the Jensen–Shannon divergence between intensity histograms~~(PDF–JS) and~~. Conv2D-MMD^2 ~~in~~ is an MMD in the feature space of a fixed random encoder~~'s feature space [11, 4]~~ ~~,~~ [11, 4]. The arms set flux in different ways, so we add a post-hoc flux-aligned Conv2D-MMD^2 ~~rescaling~~ that rescales every frame to the held-out mean total~~since the arms do not agree on how flux is set, and report a held-out~~. For unseen images we report the projection correlation, taking the best of eight latent restarts by the fit's own loss. Bins, scale and bandwidth are set on the held-out data alone, so no metric moves with the arm scored~~, and every~~. Every arm generates 200 images against the same 200 held-out frames at $N \in \{11, 100\}$~~, every number~~. Each number is a mean $\pm$ sample sd over five seeds, shown beside a real-vs-real floor between two disjoint real sets. The N training images are fixed across seeds, so ~~that~~ the spread is initialisation and sampling variance alone. A lead in sd ~~means~~ is the difference of two means in units of the baseline's seed sd, ~~descriptive~~ a descriptive quantity and not a significance test.

~~Departures from the reference setup.~~ ~~Four choices differ, each following from the data. The reference snapshots interpolate to unseen times, whereas the DeepLense `Model II` images we use [19] are static, so we generate unconditionally and project onto held-out images. Its 18-bit $(x, y, \text{intensity})$ encoding suits a field of tuples, whereas a lens crop's raw intensity axis carries barely one bit, so we centroid-align each image and read its central 32×32 crop, 99.4% of the light, as one probability mass function (pmf) over 1,024 cells at $n = 10$ (App. A). It fixes the latent block, whereas we learn the map and train the N images jointly, and it estimates its MMD from samples where a 10-qubit pmf is exact.~~

3 Results

3.1 Where lens crops sit on the correlation–complexity map

Table 1 recomputes the map (Fig. 1(a)). ~~Encoded as the reference paper encodes~~ Under the encoding the reference paper uses for turbulence, lens images sit far above ~~it~~ turbulence in I_{QCLI} and inside the corner the map calls IQP-compatible~~, yet a permutation null keeping~~. A permutation null that keeps each image's intensity histogram comes within 14% of the lens value~~and~~, and within a factor of two of the much smaller turbulence value~~, so there the index largely~~. Under this encoding the index therefore mostly tracks the histogram. Both indices depend on K, the number of codes per image, so ~~a row reads only~~ each row should be read against its own-K null~~; at~~. At the protocol's own $N_t = 5 \times 10^4$ samples per snapshot [15] the turbulence I_{QCLI} falls to 0.091, against a null of 0.098. The 10-bit crop pmf, which is the encoding we model, ~~is the one that~~ separates from its null in both coordinates~~, but~~. It sits at $I_{\mathrm{CCI}} = 0.43$, outside the high-I_{CCI} corner, so its ~~favourable~~ high I_{QCLI} motivates the encoding ~~, not the choice of~~ but gives no reason to prefer an IQP over a pairwise model.

3.2 From the reference protocol to a non-linear latent map

Run sequentially on our own 50-phase block (App. C), Algorithm 1 produces noise (Fig. 1(b), second row). ~~Neither representability nor the block is the cause, since the full circuit~~ The circuit can represent the data, since it fits a *single* aligned crop to correlation 0.99997 ± 0.00001~~and~~. The block is not the cause either, since training the same 50 phases jointly with the core across all N images (the *fixed basis, joint* row of Table 2) ~~already~~ recovers most of the gap. ~~What fails is the frozen core with sequential per-image adaptation, since with the core frozen~~ The failure lies in freezing the core on one anchor ~~the~~ and adapting each image in sequence. With the core frozen, the 50 fixed coordinates reconstruct ~~every other training image~~ the other training images at only 0.44 ± 0.04~~;~~. We did not sweep the block size of 50~~is not swept~~. Equation (1) ~~repairs the rest but introduces~~ closes the rest of the gap but has a width limit of its own. On a 20-image sweep ~~the learned linear map holds a~~ (Table 3) the held-out projection correlation of ~~0.906 ± 0.002 at M=2, falls to 0.738 ± 0.067 at M=6 and collapses to 0.218 ± 0.121 at~~ the learned linear map falls steeply from M=2 to M=8. The non-linear map ~~removes that collapse , holding 0.9368 to 0.9519 across the sweep, a little above that split's~~ avoids that collapse and stays slightly above the nearest-training-image reference~~of 0.9339; the two sweeps come from separate drivers differing in optimiser schedule and initialisation, both scored on the same 20 held-out frames, so a small offset between them is not a model difference. It~~. The non-linear map is flat to M=4 and degrades from M=8, so it does not buy a large useful width~~, being flat to M=4 and degrading from M=8. We report it at~~. At M=2, where ~~the~~ we report it, its held-out projection correlation is 0.901 at N=11 and 0.977 at N=100~~, against 0.892 and 0.970 for projecting~~. Projecting the nearest training image ~~and 0.838 and 0.887 for~~ gives 0.892 and 0.970, and a single zero-initialised restart gives 0.838 and 0.887.

3.3 Comparison with classical baselines

~~In Table 2 the eleven-image regime is where the~~ The IQP arms are competitive in the eleven-image regime of Table 2. There the non-linear arm matches the DCGAN on PDF–JS and raw Conv2D-MMD^2 within seed noise ~~(−0.13 and −0.04 of the DCGAN's own sd) and is ahead~~ and leads on the flux-aligned column~~(+1.57 sd), and against the RBM it is ahead~~. It leads the RBM on the two MMD^2 columns ~~by 1.40 and 2.27 of the RBM's sd with PDF–JS inside one sd (0.75), at~~ and is within one sd on PDF–JS (last two rows of Table 2), with a sixth of the DCGAN's parameters and three times the RBM's. Under the paper kernel the same map ~~holds the best~~ has the lowest mean PDF–JS, 0.75×10^{-3} against ~~the DCGAN's 1.1 ± 0.6, which leaves the two inside~~ 1.1 ± 0.6 for the DCGAN, still within one sd, and ~~trades that for~~ a worse flux-aligned column. ~~No arm is ahead of~~ At neither size does an arm lead the DCGAN on all three columns by more than a seed sd~~at either size, and at~~. At 100 images ~~it is ahead~~ the DCGAN leads on PDF–JS and raw Conv2D-MMD^2, and the flux-aligned 1.8 ± 0.2 ~~sitting inside~~ sits within one sd of its 2.1 ± 0.4.

~~Three properties bound how the table reads.~~ The flux-aligned column ~~is invariant to~~ does not depend on the intensity convention, and the largest paired difference ~~being 6×10^{-9}~~ over the 325 runs carrying both ~~. That is invariance, not independencefrom the raw column, so~~ is 6×10^{-9}. Invariance is not independence, and the two MMD^2 columns ~~measure shape, and~~ overlap, since the aligned one measures shape and the raw one measures shape plus flux~~, not two separate things~~. The sampling rule ~~is also not neutral between arms , improving every IQP arm's~~ also treats the arms unequally. At N=11 it improves the raw Conv2D-MMD^2 of every IQP arm by 22 to 39% ~~at N=11 while making~~ and makes the RBM's 20% worse, and under the fixed intensity the RBM leads the non-linear arm

Table 1: Correlation–complexity map ~~, each~~ over five seeds. Each permutation null ~~keeping~~ keeps the value histogram of the row above it, ~~five seeds;~~ and ± 0.0000 marks a seed spread below 5×10^{-5}.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit $(x, y, \mathrm{intensity})$ tuples	4,096	0.4715 ± 0.0000	0.8677 ± 0.0000
permutation null	4,096	0.4154 ± 0.0000	0.8783 ± 0.0001
Turbulence D2, 18-bit tuples	4,096	0.0232 ± 0.0002	0.9285 ± 0.0008
permutation null	4,096	0.0114 ± 0.0000	0.9371 ± 0.0009
Lens, 10-bit crop pmf	exact pmf	0.4731 ± 0.0000	0.4313 ± 0.0000
permutation null	exact pmf	0.1782 ± 0.0000	0.9634 ± 0.0004

Table 2: Lens crops under the sampled intensity ~~,~~over five seeds, with PDF–JS in 10^{-3} and both MMD^2 columns in 10^{-2} ~~. Lower~~ (lower is better~~; **bold** ~~). **Bold** marks a best mean whose gap to ~~*every*~~ every other arm exceeds the sum of the two seed sds, so a column without bold ~~carries~~ has no win the seed spread supports. Params are totals at $N{=}11$ / $N{=}100$~~and~~ . App. C ~~gives~~ defines † and the parameter convention. The last two rows give the lead of the non-linear flat-MMD arm at $N{=}11$ in the baseline's seed sd (Sec. 2).

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	$\mathrm{MMD}^2_{\mathrm{al}}$	PDF–JS	MMD^2	$\mathrm{MMD}^2_{\mathrm{al}}$
IQP, reference protocol (Alg. 1)†	1,524 / 5,974	6.5 ± 0.8	14.8 ± 1.6	22.5 ± 2.2	7.7 ± 0.4	14.7 ± 0.3	26.6 ± 0.7
IQP, fixed basis, joint	1,524 / 5,974	1.8 ± 0.4	8.7 ± 1.4	12.5 ± 1.3	1.1 ± 0.2	5.4 ± 0.4	8.5 ± 1.5
IQP, linear map $M{=}4$ (flat MMD)	5,164 / 5,520	2.8 ± 0.1	5.6 ± 1.9	2.7 ± 0.2	2.6 ± 0.2	2.8 ± 0.5	2.6 ± 0.4
IQP, linear map $M{=}4$ (paper kernel)	5,164 / 5,520	1.0 ± 0.05	6.3 ± 1.5	7.4 ± 0.6	1.5 ± 0.2	5.9 ± 0.4	10.0 ± 0.8
IQP, non-linear map $M{=}2$ (flat MMD)	33,910 / 34,088	1.2 ± 0.07	5.2 ± 1.8	$\mathbf{2.2} \pm 0.2$	1.7 ± 0.2	2.4 ± 0.5	1.8 ± 0.2
IQP, non-linear map $M{=}2$ (paper kernel)	33,910 / 34,088	0.75 ± 0.07	5.4 ± 1.5	5.1 ± 0.3	0.99 ± 0.04	4.2 ± 0.4	6.6 ± 0.6
RBM [29]	11,510 / 15,960	1.9 ± 1.0	7.9 ± 1.9	11.1 ± 3.9	1.1 ± 0.2	5.1 ± 0.8	8.1 ± 2.2
DCGAN [22]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	$\mathbf{0.55} \pm 0.12$	2.2 ± 0.3	2.1 ± 0.4
Real-vs-real floor	n/a	$\mathit{0.04} \pm 0.02$	$\mathit{0.5} \pm 0.3$	$\mathit{0.4} \pm 0.2$	$\mathit{0.04} \pm 0.02$	$\mathit{0.5} \pm 0.3$	$\mathit{0.4} \pm 0.2$
Lead over DCGAN, in DCGAN sd		-0.13	-0.04	$+1.57$			
Lead over RBM, in RBM sd		$+0.75$	$+1.40$	$+2.27$			

~~there~~ on that column (6.5 ± 0.8 against 8.5 ± 0.2, App. A)~~, so the~~ . The raw-column ~~standing~~ ranking against the RBM is therefore a property of the sampled decode and not of the model. Sampling ~~fixes the shape of the generated flux distribution and not its location,~~ reproduces the spread of the training totals but keeps their mean, and the eleven training frames ~~being~~ are 9% dimmer than the held-out set~~, which~~ . That offset is why the raw column ~~stops there.~~

~~Data~~ of the learned-map IQP arms at $N{=}11$ lands level with the DCGAN and not near the floor. App. B reports data scaling over $N \in \{10, \ldots, 100\}$ and per-class retraining~~are reported in App. B.~~ .

4 Discussion and limitations

~~At eleven training images a~~ The competitive standing of the 10-qubit ~~IQP Born machine with a learned non-linear latent map is competitive with an RBM and a DCGAN, matching the DCGAN on both generation metrics within the seed spread at a sixth of its parameters and holding the lowest flux-aligned mean; that standing does not survive~~ model at eleven images does not carry over to 100 images. ~~Nothing here is evidence of quantum advantage, since every circuit is simulated exactly at $n = 10$ qubits, and any deployed margin is~~ A deployed margin would also be bounded by connectivity and noise [21]. The results concern sample quality and projection fidelity alone and do not show that a quantum device would beat a classical one on this task. Generated frames sit farther in L_2 from their nearest training image than real held-out frames do (4.21 ± 0.10 against 2.90 at $N{=}11$), so the arm ~~is not memorising~~does not memorise. Two steps ~~are not shot-estimable as run, the~~ cannot be estimated from shots as run. The exact-KL warm start ~~, replaced in App. C~~ can be replaced within one seed sd ~~, and the projection , which~~ (App. C), but the projection optimises z against the exact pmf~~; and at~~ . At $n = 10$ we parameterise all 2^n Walsh phases~~where the~~ , and scaling up would need them sparsified [15], ~~untested here~~which we have not tested. Only the model fit sees N images~~, while bandwidths,~~ . Bandwidths, floors and the baselines' early stopping use hundreds to thousands of real frames, so ~~this is a small-training-set study , not a small-data pipeline; alignment~~ the study limits the training set and not the pipeline. Alignment and the crop leave translation and the outer 0.6% of the light outside the task, ~~floor included~~and the floor shares this restriction. Five seeds are few~~, the DCGAN~~ . The DCGAN alone is not bit-reproducible and adds up to 57% of spread at one seed~~and alone is not bit-reproducible, the~~ . The sampled intensity moves the raw column unequally ~~, everything~~ across arms. Everything rests on one simulated dataset, the metrics do not test substructure, and no destroyed image is scored as an upper reference.

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A Pipeline and the fixed-S convention

~~Rerunning~~ We reran Table 2 with the single training-mean intensity under which every pmf-based arm ~~published under rebuilds~~ was published. The same runs rebuild the published frames exactly~~from the same runs~~, so the two decodes are paired. ~~Sampling improves every IQP arm's~~ At $N{=}11$ sampling improves the raw Conv2D-MMD^2 ~~at $N{=}11$~~ of every IQP arm by 0.033 to 0.041 (paired sd 0.012 to 0.020) ~~while making~~ and makes the RBM worse by 0.013 ± 0.017, which ~~is why the RBM leads the non-linear arm there~~ explains the RBM lead under the fixed intensity ~~(6.5 ± 0.8 against 8.5 ± 0.2); at~~ reported in the main text. At $N{=}100$ ~~it~~ sampling improves every arm~~, the~~ . The PDF–JS effect is at most 7×10^{-4} and mixed in sign, and the flux-aligned effect is below 10^{-8}. The full table is in the released seed tables.

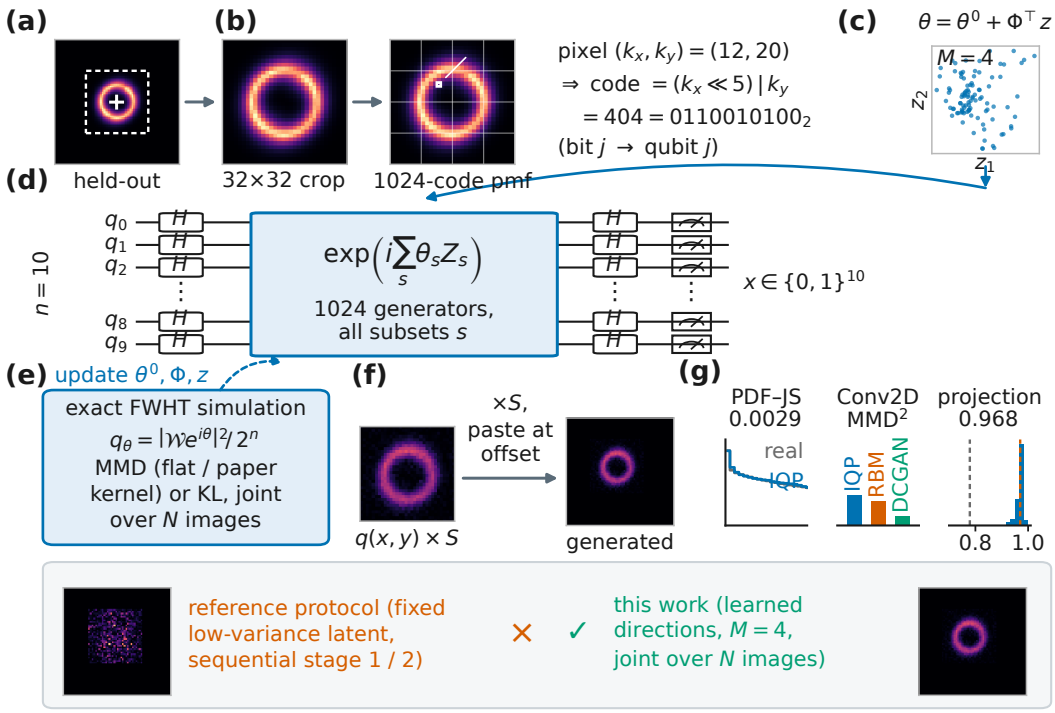


Figure 2: Pipeline. **(a)** A held-out image and the crop window. **(b)** The aligned crop as a pmf over 1,024 cells~~, the pixel~~ . Pixel $(12, 20)$ ~~carrying~~ carries code 404, and bit j ~~addressing~~ addresses qubit j. **(c)** Learned latents at $M = 4$. **(d)** The circuit. **(e)** Exact simulation, drawn for Eq. (1)~~; the~~ . The main arm holds $\theta = \theta^0 + g(z)$. **(f)** Decoding at S. **(g)** Metrics for the linear $M{=}4$ arm.

B Latent sweeps and per-class results

Hidden width matters less than latent width, ~~everything~~ since every width from $H{=}16$ up ~~being~~ lies inside a seed sd~~, and the~~ . The latent-width sweep is flat over $M \leq 4$~~before degrading~~ , degrades at $M{=}8$ and ~~collapsing~~ collapses at $M{=}16$~~; both~~ . Both full sweeps are in the released seed tables. ~~Across $N \in \{10, 30, 50, 70, 90, 100\}$ the non-linear arm's~~ In Table 4 the raw Conv2D-MMD^2 ~~falls from 0.062 ± 0.016 to 0.024 ± 0.005~~ of the non-linear arm and the RBM ~~'s from 0.091 ± 0.017 to 0.051 ± 0.008,~~ falls with N, and the DCGAN starts ahead and ends level. On the ~~DCGAN starting ahead at $N{=}10$ (0.046 ± 0.017) and ending level, while on the~~ flux-aligned column ~~it runs 0.026, 0.023, 0.020, 0.016, 0.019, 0.018 against the RBM's 0.077 to 0.127,~~ the non-linear arm is lowest at every N~~but only at $N{=}10$ and $N{=}70$~~ , but by more than a seed sd ~~. Retrained~~ only at $N{=}10$ and $N{=}70$. Retraining inside one class (Table 5, Fig. 3) shows that each class can be generated on its own~~, which is not the same as the model separating them. The~~ . It does not show that one model separates the classes. At $N{=}11$ the mean $\|z\|$ ~~at $N{=}11$~~ is 0.101 at $M{=}2$ and 0.120 at $M{=}8$, while the affine residual of g falls from 0.70 to 0.17, so g is ~~most nearly~~ closest to linear where it fails.

C Turbulence replication, training and evaluation details

On that paper's own dataset (2D cylinder-wake turbulence D2 [13], $n = 18$ qubits, 11 anchor snapshots, five seeds) ~~the result is **not reproduced**~~ we do not reproduce the result under our pre-registered convention~~, the~~ . The RBM (PDF–JS 0.217 ± 0.007) and a 1.2M-parameter DCGAN (0.045 ± 0.062) both ~~scoring~~ score better than our 31k-parameter IQP (0.272 ± 0.004) on ~~*both*~~ both metrics. Under ~~a~~ last-write-wins aggregation, ~~never specified in~~ which that paper [15] never specifies, our PDF–JS moves towards the published values without reaching them (0.0324 ~~,~~ and 0.0473 against 0.0238 ~~,~~ and 0.0221), and the RBM still leads. On these data Conv2D-MMD^2 ~~there~~ does not rank a destroyed image below a trained one~~, pixel-shuffled ground truth scoring~~ . Pixel-shuffled ground truth scores 0.7045 ± 0.0095, better than every model in that paper's Table ~~2; that paper notes itself~~

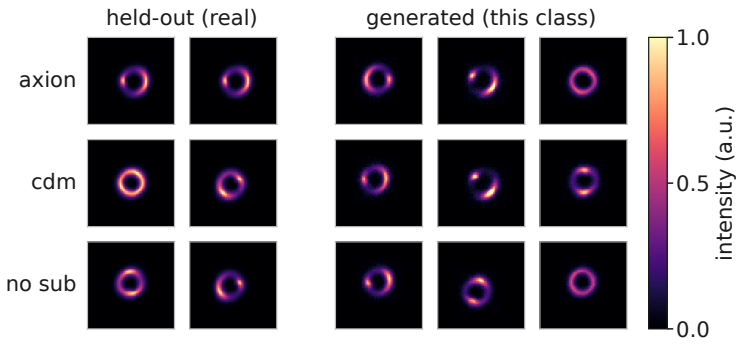


Figure 3: One row per substructure class at $N{=}100$, ~~sharing one~~ on a shared 0 to 1 intensity scale. Left, two held-out frames of that class~~; right~~. Right, three unconditional samples from the non-linear $M{=}2$ arm retrained on that class alone, taken as the first two and first three in index order ~~, no~~ without selection, seed 0.

Table 3: ~~Per class~~Projection correlation, ~~each arm retrained on that class alone and scored against that class's own 200 held-out frames; sampled intensity, five seeds, units of Table 2~~20-image sweep. ~~A column reads down a class, not across classes.~~ The ~~real-vs-real floors are 0.03/0.36/0.31 for axion~~two maps use separate drivers, ~~0.03/0.30/0.32 for cdm and 0.04/0.45/0.45 for no sub, in the same units and order~~so a small gap between them means nothing.

Latent map	M	Correlation
linear	2	0.906 ± 0.002
linear	6	0.738 ± 0.067
linear	8	0.218 ± 0.121
non-linear	all	0.9368 to 0.9519
nearest train. image	n/a	0.9339

Table 4: Data scaling over five seeds, sampled intensity, raw units. Blank cells are not reported. The RBM's flux-aligned value spans 0.077 to 0.127.

N	MMD^2 non-linear	MMD^2 RBM	MMD^2 DCGAN	MMD$^2_{\text{al}}$ non-linear
10	0.062 ± 0.016	0.091 ± 0.017	0.046 ± 0.017	0.026
30				0.023
50				0.020
70				0.016
90				0.019
100	0.024 ± 0.005	0.051 ± 0.008		0.018

2. That paper itself notes that its baselines are not exhaustive and that this metric ~~is no substitute for~~ does not replace spectral diagnostics [15].

~~Arms, baselines and protocol.~~ The † arm runs Algorithm 1 of v2 [15] on our own latent block~~,~~. That block holds the 50 subsets of Ham-

Table 6: Wasserstein-1 distance in ring radius r and ellipticity e_1, e_2. Blank cells are not reported.

Arm	$N=11$ r	$N=11$ e_1	$N=11$ e_2	$N=100$ r	$N=100$ e_1	$N=100$ e_2
multi-arm	0.82	0.37	0.15	1.21	0.11	0.14
RBM	3.23	0.46	0.24			
DCGAN	1.67	0.62	0.40	0.40	0.28	0.15
Real-vs-real	*0.10*	*0.17*	*0.10*			

Table 5: Per class, each arm retrained on that class alone and scored against that class's own 200 held-out frames, under the sampled intensity over five seeds, in the units of Table 2. Values compare within a class only. The real-vs-real floors are 0.03/0.36/0.31 for axion, 0.03/0.30/0.32 for cdm and 0.04/0.45/0.45 for no sub, in the same units and order.

		non-linear $M{=}2$			linear $M{=}4$		
Class	N	PDF–JS	MMD^2	MMD$^2_{\text{al}}$	PDF–JS	MMD^2	MMD$^2_{\text{al}}$
axion	11	1.1 ± 0.10	4.6 ± 0.6	2.9 ± 0.4	3.4 ± 0.20	4.7 ± 0.7	2.8 ± 0.5
axion	100	1.8 ± 0.17	2.9 ± 0.2	1.7 ± 0.2	2.6 ± 0.11	2.7 ± 0.2	2.1 ± 0.2
cdm	11	0.94 ± 0.05	4.4 ± 0.2	2.7 ± 0.3	2.4 ± 0.09	4.3 ± 0.3	2.6 ± 0.5
cdm	100	1.7 ± 0.06	2.1 ± 0.3	1.5 ± 0.3	2.7 ± 0.25	2.3 ± 0.4	1.8 ± 0.3
no sub	11	0.98 ± 0.10	3.1 ± 0.9	2.4 ± 0.6	2.6 ± 0.22	3.5 ± 0.8	2.6 ± 0.1
no sub	100	1.7 ± 0.10	2.3 ± 0.4	2.1 ± 0.3	2.7 ± 0.31	2.7 ± 0.4	2.9 ± 0.3

ming weight at most 2 whose Walsh coefficients vary most over a disjoint pool of 2,000 images, where the reference block is the first 50 in Hamming order~~;~~ . ~~*fixed* *Fixed* *basis, joint* ~~is that block trained~~ trains that block jointly with the core across all N images. Params are totals at that N, ~~shared~~ counting shared parameters (θ^0 included) plus N per-image latents~~, so the~~ . The fixed-basis arms hold $974 + 50N$, the learned-direction arms $5{,}120 + MN$ and the RBM $10{,}960 + 50N$~~; the DCGAN entry is~~ . The DCGAN entry counts its generator alone, ~~its discriminator adding~~ and its discriminator adds 53,497. The RBM has 10 visible and 1,000 hidden units on the 10-bit code~~, each image owning~~ . Each image owns its visible biases and the first 40 hidden biases, and the RBM ran 160k or 80k steps~~; the~~ . The DCGAN builds the crop from a 128-dimensional prior through four transposed convolutions, ran 5k to 30k steps and learns its own intensity.

The paper kernel is the three-bandwidth Gaussian MMD of [24] at $\sigma = \{0.989, 1.729, 2.236\}$~~, with~~ . The protocol we replicate publishes no bandwidths for this encoding, so we set σ_1 ~~set so~~ to make the average Pauli weight ~~is~~ 2, σ_3 to the square root of the median heuristic and σ_2 to their root mean square~~, since the protocol we replicate publishes none for this encoding~~. The Conv2D-MMD^2 bandwidth is 0.005547 raw and 0.002558 flux-aligned~~,~~ . It is fixed once on the held-out features and shared by every table except the per-class rows~~; the~~ . The 20 floor draws come from the 8,710 validation images the held-out split leaves~~; and the Silverman bandwidth [27]~~ . The bandwidth of the intensity estimate, by Silverman's rule of thumb [27], is 14.64 at N=11 and 10.36 at N=100. ~~The baselines' validation rule applied~~

Applied to the IQP arms, the baselines' validation rule selects 2k to 16k steps~~and moves every~~ . Every Table 2 cell it changes moves toward better IQP values ~~without altering a comparison~~ and no comparison flips, so our pre-registered 4,000 steps is the conservative choice~~, and at~~ . At 20,000 fixed steps both baselines are much worse, so their early stopping ~~is load-bearing and granted. Of three ablations none~~ matters and we keep it. None of three ablations moves a generation metric by more than a seed sd. A flat-MMD warm start of θ^0, ~~making~~ which makes every stage shot-estimable, leaves the flux-aligned column at 0.0231 against 0.0225 at N=11~~; picking~~ . Picking the projection restart by correlation ~~rather than by~~ instead of fit loss moves the correlation by 0.0003~~;~~ , and two restarts match eight~~(0.901, 0.977) where one zero-initialised restart reaches 0.838 and 0.887. The projection references at~~ . At M=2 ~~are 0.892 and 0.970 for the nearest training image and~~ projecting the mean crop gives 0.765 and 0.779 ~~for the mean crop~~at the two training-set sizes. Our Conv2D-MMD^2 is the kernel inception distance [4] with an untrained encoder~~, we~~ . We do not use the purpose-built generalisation metrics of [10], and classical generation of cosmological images is established [20]. ~~As Wasserstein-1 distance in ring radius and the two ellipticity components the~~ In ring morphology (Table 6) the main arm is ~~at 0.82/0.33/0.35 at~~ closest in radius at N=11 ~~against 3.23/0.46/0.24 (RBM), 1.67/0.62/0.40 (DCGAN) and 0.10/0.17/0.10 between two real sets; at N=100~~ and the DCGAN is closest ~~, 0.40/0.28/0.15 against 1.01/0.31/0.14~~at N=100. Retraining the same DCGAN configuration at the same seed moves its raw Conv2D-MMD^2 by 5 to 57% and its PDF–JS by 2 to 74%, ~~which is what we mean by~~ and in this sense it is the one arm that is not bit-reproducible.

D Qubit count, the reference paper's Fig. 4, and the departures in full

The qubit count has to match the effective resolution of the image, a choice the reference paper does not face ~~at $n = 18$ on~~ with a field of tuples at $n = 18$. At $n = 8$ the 16×16 pmf averages the ring away and the fixed bilinear upsampling cannot ~~put itback~~restore it, so the flux-aligned error is more than twice the $n = 10$ value~~; at~~ . At $n = 12$ about three quarters of the 4,096 cells carry background rather than lens light~~, the~~ . The arm learns that background and becomes unstable across seeds, and its N=100 PDF–JS ~~reaching~~ reaches 92 ± 71~~with a spread as large as the mean~~. We therefore work at $n = 10$.

~~Re-running~~ We re-ran that paper's Fig. 4 over five ~~SPSA seeds , its positive low~~ seeds of SPSA (simultaneous perturbation stochastic approximation, which that paper uses to spread its targets over I_{QCLI}~~coupling~~). Its positive coupling at low I_{QCLI} ($I_{\mathrm{QCLI}} \leq 0.3$) reproduces at every size and

Table 7: Qubit ladder at $N{=}11$ ~~,~~over five seeds, in the units of Table 2. The flux-aligned column ~~is invariant to~~ does not depend on the intensity convention (App. A) and compares across rows~~; the~~. The other two ~~are under~~ columns use the fixed intensity, except in the $n = 8$ row ~~,~~(marked *). The latent map differs by row, so ~~this~~ the table is a ladder of encodings ~~,~~and does not ~~a sweep of~~ vary n alone.

n	Encoding	Latent map	Params	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$
8	16×16 pmf, bilinear upsampling	MLP $H{=}8$, $M{=}2$	2,350	$1.7 \pm 0.05^*$	$8.0 \pm 1.7^*$	5.1 ± 0.3
10	32×32 aligned crop pmf	MLP $H{=}32$, $M{=}2$	33,910	1.0 ± 0.1	8.5 ± 0.2	$\mathbf{2.3} \pm 0.2$
12	whole 64×64 frame as one pmf	linear, $M{=}4$	20,524	43 ± 30	23.3 ± 8.8	18.4 ± 11.3

in every seed, with Spearman $+0.92 \pm 0.10$, $+0.91 \pm 0.04$ and $+0.96 \pm 0.01$ at $n = 8$, 12, 16~~, while above~~. Above that range the sign of the global coefficient changes from seed to seed.

Table 8: ~~Every place our protocol departs~~ Each departure from the ~~one~~ protocol we replicate, with the property of the data that drove it~~; none~~. None of them ~~is a~~ came from tuning~~result~~.

	Reference protocol	This paper	Property of the data
Encoding	18-bit $(x, y, \text{intensity})$ tuples, many per snapshot	one 10-bit pmf over 1,024 cells of an aligned 32×32 crop	the raw intensity axis carries barely one bit and 99.4% of the light is inside the crop
Latent	a fixed block of 50 phases, the first in Hamming order	a learned map, linear $\Phi^\top z$ or a $\tanh$ $g(z)$	on static images fixed directions cross-fade pixels instead of moving ring structure
Training	core fitted on one anchor, then each sample adapted in sequence	core and all N latents trained jointly, 4,000 Adam steps	with at most 100 static frames there is no stream to adapt along
Objective	multi-bandwidth Gaussian MMD estimated from samples	flat MMD as the main arm, that kernel kept as a secondary one	at $n = 10$ the pmf is exact, so sampling noise would be a choice
Flux	every decoded frame carries the training-set mean total	S drawn per image from a density estimate of the N totals	real crops spread by about a quarter of their mean, which one S cannot reproduce
Evaluation	held-out times, that paper's own metrics	200 held-out frames fixed across arms and seeds, bins and bandwidth set on them alone	the images are static, so time cannot be held out
Baselines	compared at a fixed budget	RBM and DCGAN early-stopped on 200 validation frames, IQP arms not	the baselines converge at very different step counts, so one budget would favour us